

Campus Connect ERP

Integrated Campus Management System

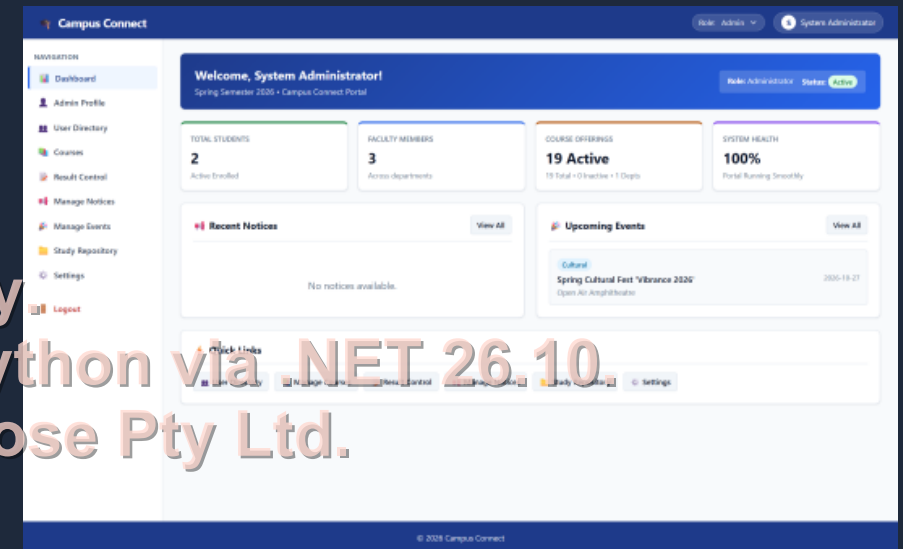
A centralized, role-based college portal for real-time attendance, 4-tier examination marks entry, automated grade calculation, and coursework management.

Evaluation only
Created with Aspose.Slides for Python via .NET 26.10.
Copyright 2004-2026 Aspose Pty Ltd.

PROJECT OVERVIEW & TECHNOLOGY STACK

- Core Technologies: Python 3.11 • Flask Web Framework • PostgreSQL • SQLAlchemy 2.0 ORM
- Developer Attribution: Ayan Mulani (Project Lead) & Project Team | SITCOE Yadrav
- GitHub Repository: <https://github.com/ayanmulani4867-cyber/Campus-mini2>

AUTHENTIC SYSTEM DASHBOARD INTERFACE



Verified System Architecture:

- ✓ Multi-Tier Role Governance: Admin, Faculty, and Student
- ✓ Live Cloud Database: Relational PostgreSQL Schema
- ✓ Demonstration Dataset: Test environment sample records

Problem Statement & Project Objectives

OPERATIONAL CHALLENGES (PROBLEM)

• **Fragmented Academic Records**

Student profiles, course lists, and marks reside in disconnected physical registers and spreadsheets, causing data silos and duplication.

• **Manual Attendance & Grade Tabulation**

Paper-based roll-calls cause administrative delays, calculation errors, and slow identification of students falling below attendance thresholds.

• **Absence of Role-Based Workspace Isolation**

Unsecured access without strict Role-Based Access Control (RBAC) creates data privacy risks between administrators, faculty, and students.

CAMPUS CONNECT ERP (SOLUTION)

✓ **Centralized Relational Database**

Unified PostgreSQL repository managed via SQLAlchemy 2.0 ORM, integrating students, faculty, departments, and course offerings.

✓ **Automated 4-Tier Evaluation Engine**

Structured scoring for CA1 (20), CA2 (20), Mid-Sem (30), and End-Sem (70) with automated SGPA/CGPA grade card generation.

✓ **Strict Role-Based Access Control (RBAC)**

Server-side session authentication with token isolation guaranteeing dedicated, secure workspaces for Admins, Faculty, and Students.

System Architecture & Data Flow Diagram



CLOUD HOSTING & DEPLOYMENT PIPELINE

- **Render Cloud Web Service:** Automated continuous deployment configured and synchronized with the repository's main branch.
- **Gunicorn WSGI Production Server:** Multi-worker Python WSGI server handling concurrent HTTP requests with ProxyFix middleware.
- **Zero-Mock Relational Integrity:** Every application route executes authentic SQL queries against PostgreSQL tables; no hardcoded dummy data.

Administrative Dashboard & User Directory

Campus User Directory (Admin Management)
Centralized database of registered students, faculty members and staff

ENROLLED STUDENTS 2
Undergraduate & Postgrad

TEACHING FACULTY 3
Permanent & Visiting

ENGINEERING DEPTS 6
Academic Departments

DIRECTORY HEALTH 100%
All records synced

Search by name, roll, PRN, employee ID, email or user ID

All Semesters All Divisions All Roles All Departments

Apply Filters Reset

Campus Members Database
Showing 6 of 6 registered members

Sort by: Name

MEMBER ID / PRN	FULL NAME & EMAIL	ROLE	DEPARTMENT	CLASS / DESIGNATION	STATUS	AUTHORIZED ACTIONS
24101099	Aarav Sharma student.test@sitcoe.ac.in	Student	Computer Science & Engineering	TY B.Tech - Sem 6 - Div A	Active	View Grade Card Delete
24101005	Ayan Mulani ayanmulani4867@gmail.com	Student	Computer Science & Engineering	3rd Year - Sem 5 - Div B	Active	View Grade Card Delete
FAC-UNAUTH	Prof. Other Faculty unauth.fac@sitcoe.ac.in	Faculty	Computer Science & Engineering	Assistant Professor	Active	View Delete
FAC-TEST-01	Prof. Test Faculty prof.test@sitcoe.ac.in	Faculty	Computer Science & Engineering	Assistant Professor	Active	View Delete
FAC2026001	Rahul Sharma rahul@gmail.com	Faculty	Computer Science & Engineering	Associate Professor	Active	View Delete

System Administrator

Figure 1: Campus User Directory — Centralized account governance & profile linking
Demonstration Dataset: Role filtering (Admin, Faculty, Student) • Department search controls

VERIFIED CAPABILITIES

✓ Unified Member Ledger

Directly couples login credentials with student and faculty academic records to prevent orphan data.

✓ Multi-Field Search & Filters

Search registered members by name, roll, PRN, or employee ID; filter by department and semester.

✓ Role-Based Governance

Administrators manage account activation, inspect profiles, and trigger instant student grade cards.

* Screen captured from running ERP using demonstration test dataset.

Academic Management: Attendance & Examination Results

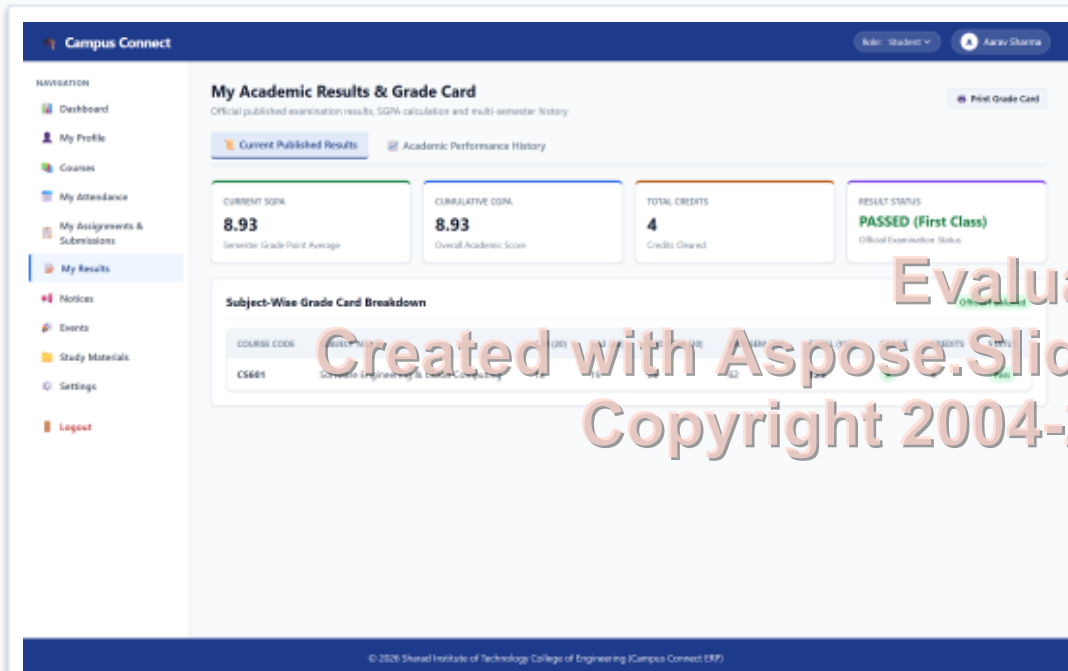


Figure 2: Published Student Grade Card with SGPA & CGPA Calculation

Sample Record: CA1 (18) + CA2 (19) + Mid (26) + End (62) = 125/140 Marks • Grade A • SGPA 8.93

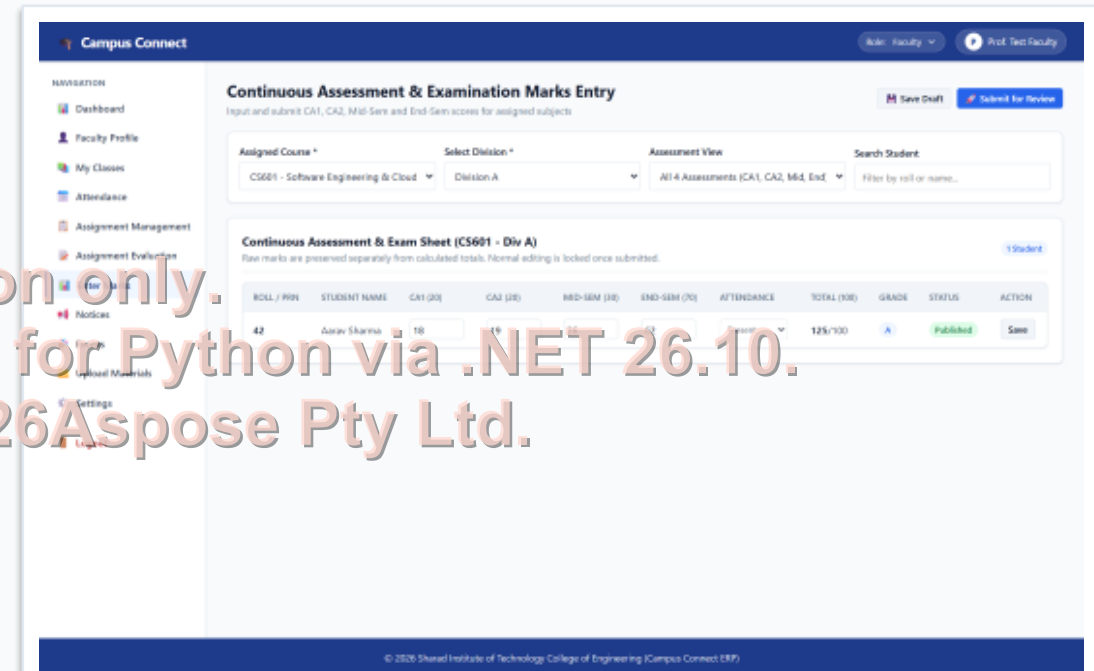


Figure 3: Faculty Continuous Assessment Sheet (CA1, CA2, Mid-Sem, End-Sem)

Workflow: CS601 Software Eng (Div A) • Draft Saving → Review Submission → Approval

MATHEMATICAL CALCULATION & VERIFIED ASSESSMENT RULES:

- 4-Tier Assessment Math: CA1 (20) + CA2 (20) + Mid-Sem (30) + End-Sem (70) = 140 Total Maximum Marks. Obtained: 125/140 = 89.29% (Grade A, Passed First Class).
- Attendance Roll-Call Ledger: Course-wise session tracking (CS601, ML, Software Testing) with automated 75% university eligibility threshold monitoring.

Coursework Lifecycle & Campus Announcements

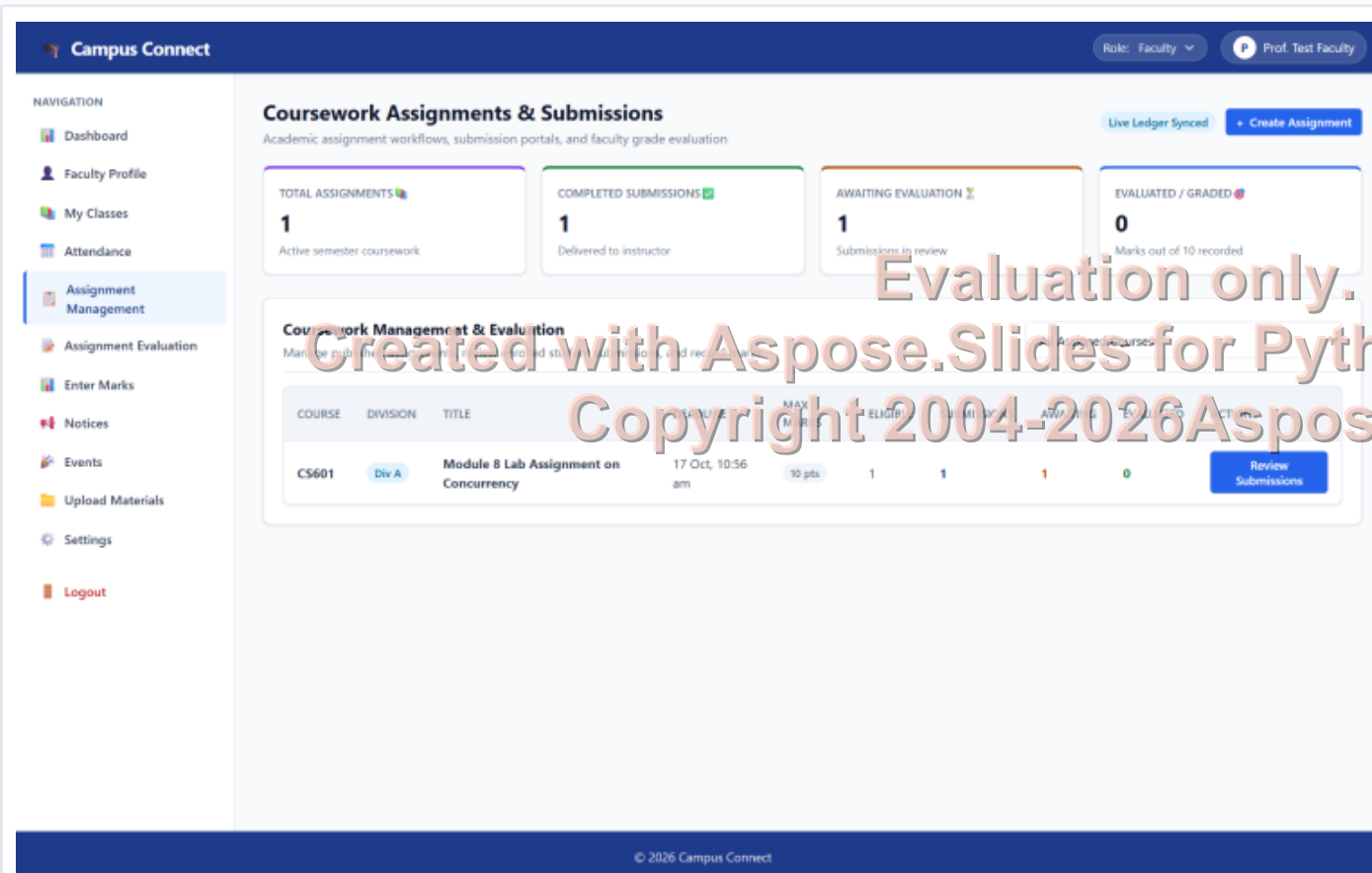


Figure 4: Coursework Assignment Management with submission locking & evaluation queue
Active Coursework: CS601 'Module 8 Lab Assignment on Concurrency' (10 pts) • 1 Delivered Submission

COLLABORATION WORKFLOWS

✓ Assignment Publishing

Faculty create course assignments with point allocations, division targets, and strict submission deadlines.

✓ Tamper-Proof Locking

Students upload solutions via their portal; submissions are timestamped and locked against post-deadline editing.

✓ Zero-Mock Circulars

Official notices and campus events (e.g. Spring Cultural Fest 'Vibrance 2026') served directly from PostgreSQL.

* All modules query genuine database tables in the demonstration environment.

Project Outcomes, Roadmap & Conclusion

🎯 THREE CORE PROJECT OUTCOMES

✓ Unified Academic Operations

Successfully integrated roll-call attendance, 4-tier continuous examination assessment, and official grade card publication into a single platform.

✓ Strict Role-Based Security

Enforced Role-Based Access Control (RBAC) with session token isolation, providing secure, partitioned workspaces for Administrators, Faculty, and Students.

✓ Production Relational Architecture

Implemented a robust PostgreSQL schema via SQLAlchemy 2.0 ORM with complete referential integrity and zero reliance on mock/demo data fallbacks.



TWO PLANNED FUTURE ENHANCEMENTS

★ Automated Parent SMS Alerts (Planned)

Integrate an SMS gateway (Twilio / Fast2SMS) to automatically alert parents and students when attendance drops below the 75% university eligibility threshold.

★ Accreditation & Analytics Reporting (Planned)

Build graphical department-wise analytics and automated PDF/Excel report exports for NAAC and NIRF institutional accreditation reviews.

CAMPUS CONNECT ERP • SITCOE YADRAV • THANK YOU!

GitHub Repository: <https://github.com/ayanmulani4867-cyber/Campus-mini2> • Open for Viva Questions & Discussion